LABORATORY MANUAL

SC4172
IoT - TinyML

Lab Experiment #5

Interfacing of Camera Module with Nano 33 Sensor board

COLLEGE OF COMPUTING AND DATA SCIENCE
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1. OBJECTIVES

This practical exercise is for you to get familiarized with the interfacing of the OV5642 camera module with the Arduino Nano 33 BLE Sense board, such that you can use them to further develop computer vision based ML applications.

2. LABORATORY

This experiment is conducted at the **Software Laboratory 1 (Location: N4-01a-02)**.

3. HARDWARE EQUIPMENT

- Computer installed with Anaconda and Tools for TensorFlow ML.
- Arduino Nano 33 BLE Sense board
- OV5642 Camera
- Arduino IDE installed with TensorFlow Lite (TFLite) library

4. REFERENCES

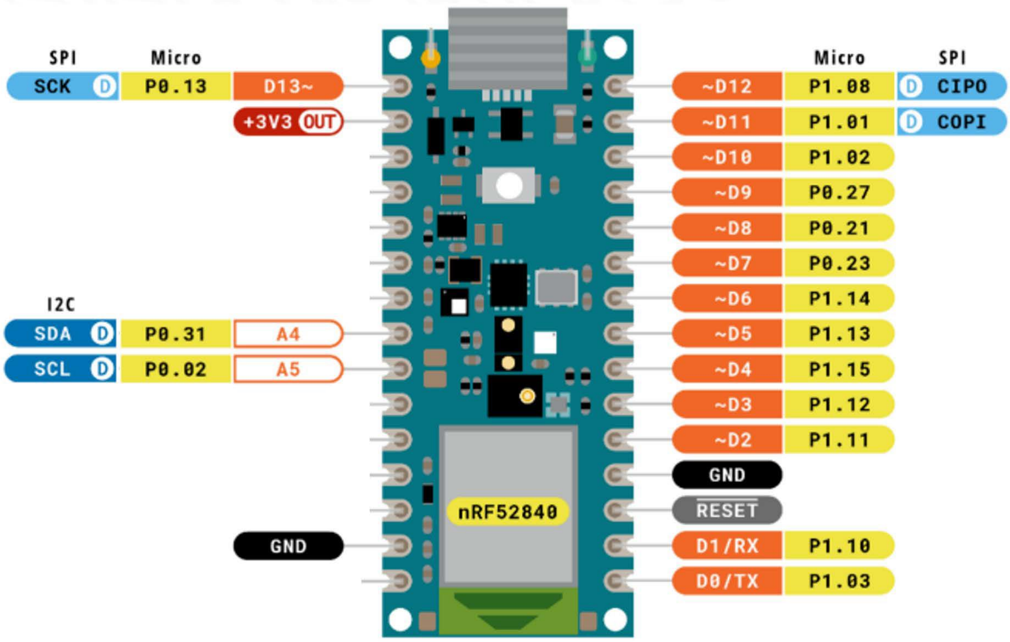
- TinyML book by Pete Warden, Daniel Situnayake. 2019. https://www.oreilly.com/library/view/tinyml/9781492052036/
- Nano33 BLE Sense Board. https://docs.arduino.cc/hardware/nano-33-ble-sense/
- SC4172 Lecture Notes

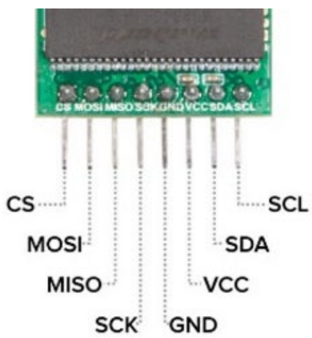
5. INTRODUCTION

The OV5642 camera is a module supported by the ArduCAM project, an open source project for Arduino camera. As such the driver library required is available through its website (https://github.com/ArduCAM/Arduino).

6. EXERCISE

Connect the OV5642 camera module to the Nano 33 Sense board using both its SPI and I2C interfaces as shown below. (I2C is needed for the board to configure the camera module)





- Download the latest ArduCAM package from https://github.com/ArduCAM/Arduino, install the necessary files and programs to see that the interfacing works as expected.
- **Watch the video** on the ArduCam setup and sample application. Video is located further down the github site.

Arducam MINI Camera Demo Tutorial for Arduino

- Overview of steps outlined in the video
  - Unzip the code and copy the ArduCam and two other directories to the **Arduino Library folder**.
  - Modify **memorysaver.h** to configure for the correct camera module. The camera module used in Lab5 is the **OV5642_MINI_5MP_PLUS**.
  - Compile and upload the sample application for OV5642 (**ArduCAM_Mini_5MP_OV5642_Plus_Functions**) to **nano33 sense** hardware.
  - Execute the **ArduCAM_Host_V2.exe** host application in Windows/Mac. The Mac and Linux version of HostApp is located at the end of the GitHub site.

5. How to download the Host V2 ?

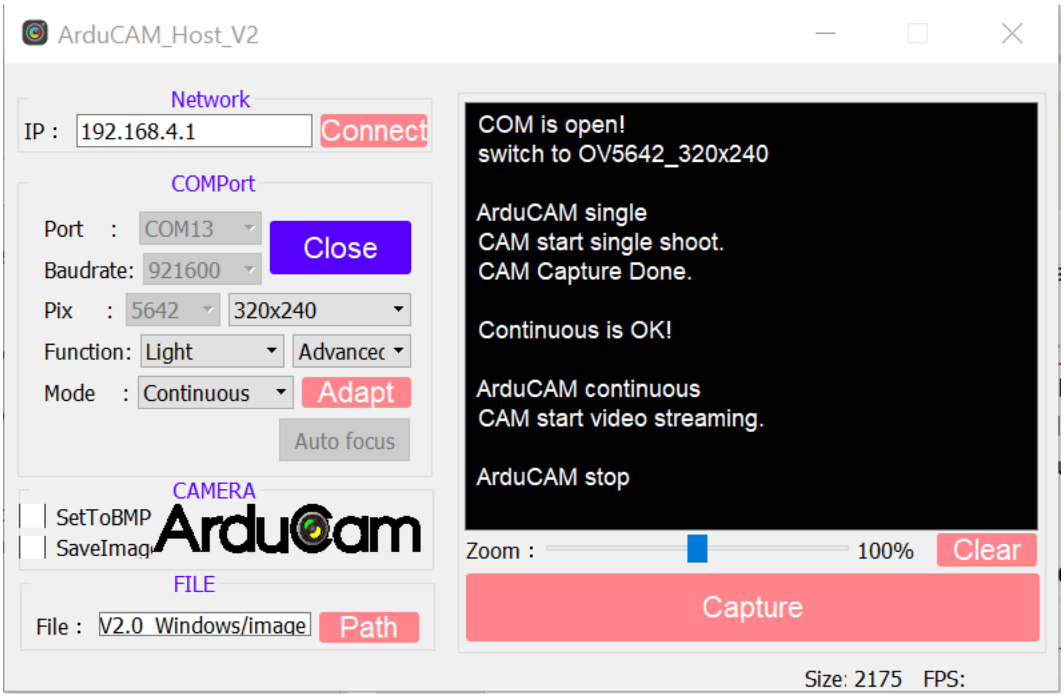
- For ArduCAM_Host_V2.0_Mac.app, please refer to this link:

```
www.arducam.com/downloads/app/ArduCAM_Host_V2.0_Mac.app.zip
```

- For ArduCAM_Mini_V2.0_Linux_x86_64bit, Please refer to this link:

```
www.arducam.com/downloads/app/ArduCAM_Mini_V2.0_Linux_x86_64bit.zip
```

  - Configure the Host_V2 app for the correct camera and other parameters.



  - Try both the single and continuous mode. Mine only works in continuous mode, hopefully your single mode works as well.

- There are some feedback that the following SPI interface speed result in more stable operation.
  - SPI.beginTransaction(SPISettings(4000000, LSBFIRST, SPI_MODE0));